

# **The Geometry of Reduction**

Sergey Omarov\*

April 2026

### **Abstract**

The document describes the structural relation between the full arrangement of what is and that slice of it which is accessible to observation. It formulates this relation in general terms, without attachment to any concrete physical, philosophical, or psychological content. The geometry of reduction describes four levels of structural relation: between  $S$  and  $R(S)$ , the division of components into symmetric and asymmetric, the parametric space of points of reduction with a subset of points allowing an observer, and the double nesting of reductions — the external  $S \rightarrow R(S)$  and the internal one, structuring separate dimensions of  $R(S)$  through the observer's memory.

## Contents

|                                                             |   |
|-------------------------------------------------------------|---|
| 1. The nature of these materials . . . . .                  | 3 |
| 2. The full structure and the reduced description . . . . . | 3 |
| 3. Symmetric and asymmetric components . . . . .            | 4 |
| 4. The space of points of reduction . . . . .               | 4 |
| 5. The observer . . . . .                                   | 5 |
| 6. Internal reduction . . . . .                             | 5 |
| 7. The structurable dimension . . . . .                     | 5 |
| 8. The limit of descriptive means . . . . .                 | 6 |
| 9. Summary . . . . .                                        | 6 |

### 1. The nature of these materials

The present document describes the structural relation between the full arrangement of what is and that slice of it which is accessible to observation. It formulates this relation in general terms, without attachment to any concrete physical, philosophical, or psychological content. Concrete applications are possible, but do not enter into the present text.

---

### 2. The full structure and the reduced description

The full structure  $S$  is the arrangement of what is, taken in its entire composition.

The observed description is not  $S$ , but its reduced form  $R(S)$ , where  $R$  is the reduction operator, transforming the full structure into a form suitable for registration through observation.

Reduction by its very meaning is a transformation with losses. If there are no losses, then it is not reduction, but a full transformation. Therefore, between  $S$  and  $R(S)$  there always exists a difference, and this difference is constitutive for the very notion of reduction.

This proposition does not introduce new entities. It fixes a structural relation that holds whenever the observed is separated from the full arrangement of the observed.

---

### 3. Symmetric and asymmetric components

Components of  $S$  symmetric relative to  $R$ , under reduction, yield stable forms. These forms reproduce themselves identically under every application of  $R$  and constitute that which in  $R(S)$  is read as laws, regularities, invariants.

Components of  $S$  asymmetric relative to  $R$ , under reduction, yield residues — traces not reducible to stable forms. In  $R(S)$  they are read as anomalies, residual tensions, parameters inexplicable within the framework of the derived laws.

The difference between the symmetric and the asymmetric part of  $S$  is not a difference in their own nature. It is a difference relative to  $R$ . One and the same component of  $S$  may be symmetric relative to one reduction and asymmetric relative to another.

Therefore, the composition of laws and the composition of anomalies in  $R(S)$  is determined not only by  $S$ , but also by the choice of  $R$ .

---

### 4. The space of points of reduction

$R$  is not a unique transformation. The reduction operator has a parametrization — a multitude of possible ways to carry out the reduction from  $S$ . To each value of the parameter there corresponds its own point of reduction, and in each such point its own  $R(S)$  arises.

The space of points of reduction is a parametric space, not a space of actualizations. The points in it have the status of possibilities. A statement about this space is a statement about the structure of  $S$ , allowing different ways of reduction, and not a statement about the existence of a multitude of separate worlds.

At each point of reduction the composition of symmetric and asymmetric components is its own. Therefore, both the composition of laws and the composition of anomalies in  $R(S)$  differs from point to point.

The content of a concrete point of reduction — how exactly  $R(S)$  looks in it — is not derived from the geometry itself. The geometry describes the structure of the relation between  $S$  and the point, but does not prescribe with what precisely the point is filled.

---

## 5. The observer

$R(S)$  acquires definiteness as a describable structure only in the presence of an observer at the point.

The observer is that through which  $R(S)$  becomes not merely a potential form, but a structure having content. Without an observer, the point of reduction remains a point in the parametric space, but does not generate a describable slice of what is.

Not every point of reduction admits the formation of an observer. The subset of points in which an observer is possible is a proper subset in the space of all points of reduction.

This subset is not singled out by external selection. It is a consequence of which points are compatible with the assembly of a structure capable of acting as an observer.

---

## 6. Internal reduction

At a point admitting an observer, in addition to the external reduction  $S \rightarrow R(S)$ , an internal reduction operates, through which separate dimensions of  $R(S)$  acquire their structure.

The internal reduction performs, with respect to one dimension of  $R(S)$ , the same function which the external reduction performs with respect to  $S$  as a whole: it transforms material into a structured form with losses.

The bearer of the internal reduction is the function of retention and succession — memory. Memory does not create the dimension to which it is applied. It structures it.

It follows from this that at a point admitting an observer there exist at least two nested reductions: the external, separating  $R(S)$  from  $S$ , and the internal, forming the structure within  $R(S)$  through the work of the observer's memory.

---

## 7. The structurable dimension

Time, in the  $R(S)$  of a point admitting an observer, is a dimension acquiring its structure through the work of the internal reduction.

Time as material may be present in  $R(S)$  as one of its components. The structure of time — sequence, directedness, the distinction of earlier and later — is not present in the material by itself. It arises through the memory of the observer.

Therefore, the statement about the structuredness of time pertains to a point admitting an observer, and not to  $S$  as such. At points not admitting an observer, there is no ground for ascribing structure to time, and therefore no ground for speaking about temporal unfolding as such.

This is not a statement about the existence at unnameable points of structureless time. This is a statement that without a structuring function the category of time in the full sense is not applicable to the point.

---

## **8. The limit of descriptive means**

In  $R(S)$  there may be present entities not reducible to its descriptive means.

Such an entity is registered in  $R(S)$  — it is present in it, interaction with it is possible, its properties are amenable to measurement. But certain aspects of its arrangement are not legible in the categories of  $R(S)$ .

An empirical example is light. It is present in the  $R(S)$  of our point, is registered and measured. At the same time, certain of its properties — in particular those which in our  $R(S)$  are taken as zero or as boundary values — indicate that light does not fit into our  $R(S)$  in full.

This is not a statement about light's own arrangement. This is a statement about the limit of  $R(S)$ : there is in it something going beyond its descriptive means.

It follows from this that  $R(S)$  is not closed relative to its own content. It may include elements whose arrangement is not derivable from its internal categories.

---

## **9. Summary**

The geometry of reduction describes four levels of structural relation.

The first — the relation of  $S$  and  $R(S)$ : the full structure and its reduced description, always separated by a constitutive difference.

The second — the division of components of  $S$  into symmetric and asymmetric relative to  $R$ , yielding in  $R(S)$  laws and residues respectively.

The third — the parametric nature of  $R$ , generating a space of points of reduction, in which a subset of points admits an observer.

The fourth — the double nesting of reductions at a point admitting an observer: the external reduction  $S \rightarrow R(S)$  and the internal reduction, structuring separate dimensions of  $R(S)$  through the memory of the observer.

Within  $R(S)$  there may be present entities not reducible to its descriptive means. This fact does not violate the geometry, but indicates its own boundary:  $R(S)$  describes that which fits into it, and does not lay claim to that which does not fit into it.

---